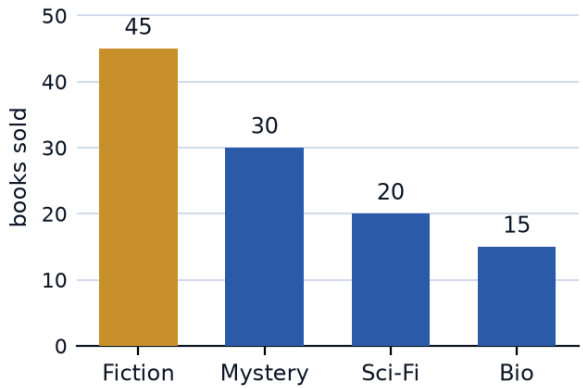
Data Analysis & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

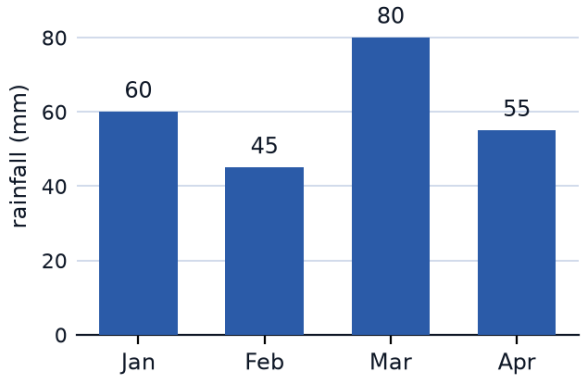
1 [Foundational] • Target: Reading bar charts

The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.



2 [Foundational] • Target: Reading bar charts

The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.



3 [Foundational] • Target: Mean, median, mode and range

Find the mean of $6, 9, 12, 15, 18$.

4 [Foundational] • Target: Mean, median, mode and range

Ages of 7 students: $13, 15, 13, 16, 14, 13, 17$. Find the mean, median, and mode.

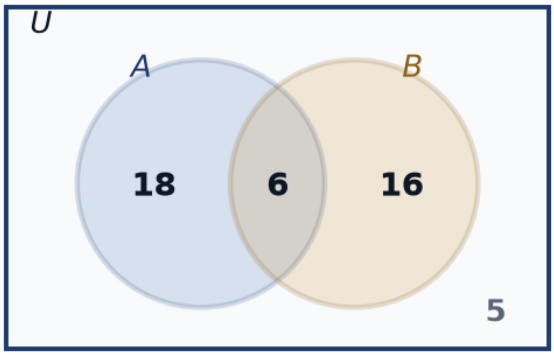
5 [Foundational] • Target: Mean, median, mode and range

Find the range and mode of the data set $8, 3, 8, 15, 8, 22, 3$.

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6 [Foundational] • Target: Mean, median, mode and range

A data set has mean 18 and 6 values. Find the sum of the values.



7 [Foundational] • Target: Mean, median, mode and range

The data set $10, 12, 11, 13, 50$ contains an outlier. Find the mean before and after removing 50.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Mean, median, mode and range

A data set of 5 values has mean 20. Four of the values are $15, 18, 22, 25$. Find the fifth value.

9 [Intermediate] • Target: Reading tables

A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?

10 [Intermediate] • Target: Reading tables

Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?

11 [Intermediate] • Target: Reading tables

A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.

12 [Intermediate] • Target: Reading tables

Using the same quiz data, find the mean score to two decimal places.

13 [Intermediate] • Target: Reading bar charts

Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?

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14 [Intermediate] • Target: Reading bar charts

Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Word Problems & Reasoning

A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.

16 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Arithmetic & Number Sense

A fair six-sided die is rolled. Find $P(\text{a number greater than } 4)$.

17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.

18 [Challenging] • Target: Basic probability • Cross-Lesson: 3.01 Algebra Essentials

A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.

19 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 4.01 Geometry & Measurement

A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.

20 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 5.01 Quantitative Comparisons

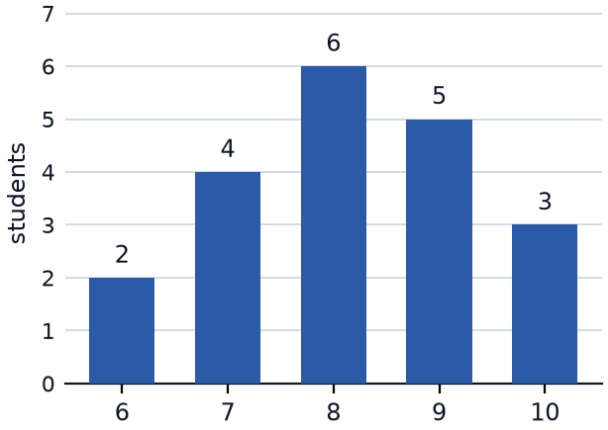
Two fair coins are tossed. Find $P(\text{both land heads})$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: MCQ — Mean, median, mode and range

The bar chart shows quiz scores. Find the total number of students, and the modal score.



a. 20 students; mode $= 8$

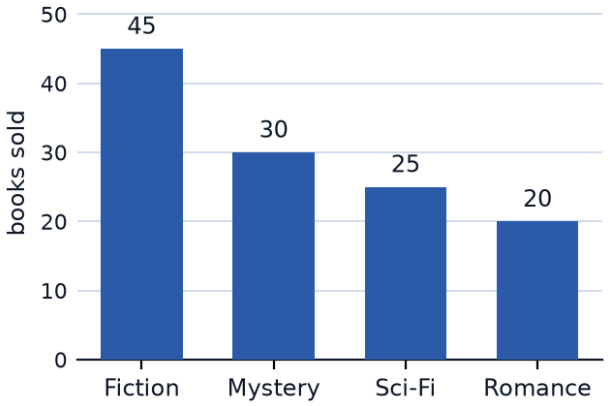
b. 20 students; mode $= 6$

c. 20 students; mode $= 9$

d. 18 students; mode $= 8$

22 [Foundational] • Target: MCQ — Reading bar charts

The bar chart shows books sold by genre in one week. Find the total books sold, and the modal genre.



a. 120; Romance

b. 100; Fiction

c. 120; Mystery

d. 120; Fiction

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23 [Foundational] • Target: MCQ — Reading bar charts

The bar chart shows quarterly sales for four stores, in thousands of riyals. Find the range, and which store had the highest sales.



a. 55; Store B

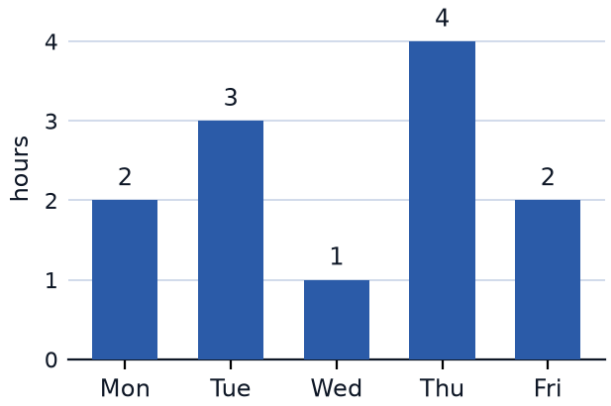
b. 55; Store D

c. 140; Store D

d. 35; Store D

24 [Foundational] • Target: MCQ — Reading bar charts

The bar chart shows hours studied each weekday. Find the total hours, and the mean hours per day.



a. 12; 2

b. 12; 2.4

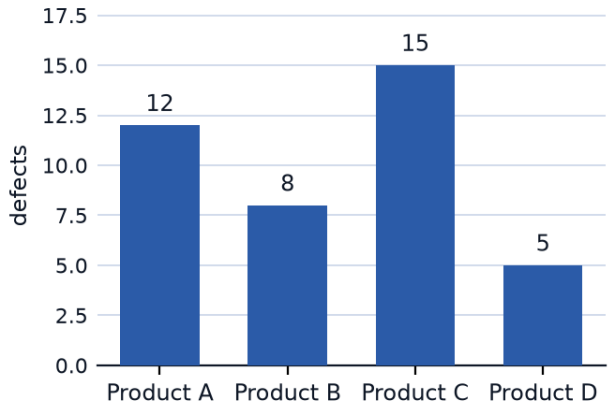
c. 12; 3

d. 10; 2

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25 [Foundational] • Target: MCQ — Reading bar charts

The bar chart shows the number of defective items found for four products in a quality check. Find the total defects, and the range.



a. 40; 10

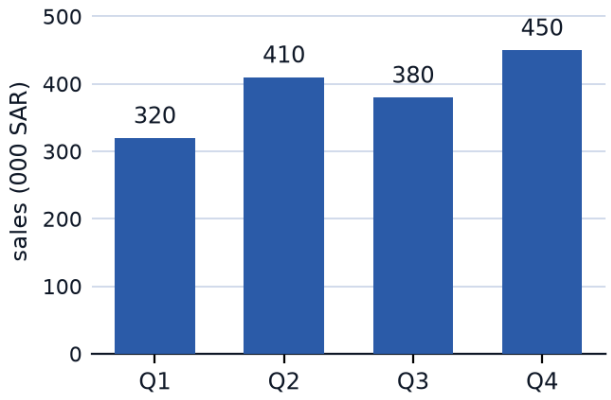
b. 40; 7

c. 35; 10

d. 40; 15

26 [Foundational] • Target: MCQ — Word problems

The bar chart shows a company's quarterly sales, in thousands of riyals. Find the total annual sales, and the mean quarterly sales.



a. 1,560; 450

b. 1,560; 390

c. 1,560; 400

d. 1,200; 390

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27 [Foundational] • Target: MCQ — Mean, median, mode and range

Find the mean of $8, 11, 14, 17, 20$.

a. 12

b. 17

c. 70

d. 14

28 [Foundational] • Target: MCQ — Mean, median, mode and range

A data set of 6 values has mean 25. Five of the values are $20, 22, 28, 30, 32$. Find the sixth value.

a. 18

b. 28

c. 8

d. 150

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Reading tables

A table shows the number of pets owned by 25 families: 0 pets (6 families), 1 pet (10), 2 pets (7), 3 pets (2). Find the median number of pets.

a. 0

b. 1.2

c. 2

d. 1

30 [Intermediate] • Target: MCQ — Basic probability

A card is drawn from a standard deck of 52 cards. Find $P(\text{a king})$.

a. $\frac{1}{4}$

b. $\frac{1}{52}$

c. $\frac{1}{13}$

d. $\frac{4}{13}$

31 [Intermediate] • Target: MCQ — Combined and complementary events

Two fair coins are tossed. Find $P(\text{at least one tail})$.

a. $\frac{3}{4}$

b. $\frac{1}{3}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

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32 [Intermediate] • Target: MCQ — Weighted averages

A class of 12 students has an average score of 78. A 13th student joins with a score of 91. Find the new average.

a. 84.50

b. 78.00

c. 91.00

d. 79.00

33 [Intermediate] • Target: MCQ — Weighted averages

Two classes combine: Class A (10 students, mean 60) and Class B (n students, mean 90), giving a combined mean of 78 for all students. Find n.

a. 12

b. 15

c. 10

d. 20

34 [Intermediate] • Target: MCQ — Comparing data sets

Two data sets both have median 20. Set P: 10, 15, 20, 25, 30. Set Q: 18, 19, 20, 21, 22. Which set is more spread out?

a. Set P

b. Cannot be determined

c. Set Q

d. They are equally spread out

35 [Intermediate] • Target: MCQ — Word problems

A survey of 150 shoppers found 90 prefer online shopping and the rest prefer in-store. Find the probability, as a fraction in simplest form, that a randomly chosen shopper prefers in-store.

a. $\frac{2}{5}$

b. $\frac{90}{150}$

c. $\frac{3}{5}$

d. $\frac{3}{50}$

36 [Intermediate] • Target: MCQ — Word problems

Find the mean of 24, 23, 48, 18, 12.

a. 30

b. 25

c. 23

d. 27

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37 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the mean of $50, 50, 43, 21, 21$.

a. 39

b. 42

c. 35

d. 37

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Algebra Essentials

Find the mean of $55, 7, 53, 41, 49$.

a. 39

b. 46

c. 41

d. 43

39 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Geometry & Measurement

A bag holds 8 red and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{8}{17}$

b. $\frac{10}{17}$

c. $\frac{9}{17}$

d. $\frac{10}{19}$

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Quantitative Comparisons

A bag holds 11 red and 5 blue marbles. Find $P(\text{red})$.

a. $\frac{5}{16}$

b. $\frac{3}{4}$

c. $\frac{13}{18}$

d. $\frac{11}{16}$

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Sequences & Patterns

A bag holds 11 red and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{13}{22}$

b. $\frac{9}{20}$

c. $\frac{11}{20}$

d. $\frac{3}{5}$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Word Problems & Reasoning

Class A has 26 students with mean score 65; Class B has 22 students with mean score 89. Find the overall mean of all 48 students.

a. 78

b. 76

c. 77

d. 74

43 [Challenging] • Target: Mean, median, mode and range • Cross-Lesson: 1.01 Arithmetic & Number Sense

Event X has probability $\frac{1}{7}$ and independent event Y has probability $\frac{4}{5}$. Find $P(X \text{ and } Y)$.

a. $\frac{1}{70}$

b. $\frac{4}{35}$

c. $\frac{33}{35}$

d. $\frac{3}{14}$

44 [Challenging] • Target: Reading tables • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Find the mean and median of $27, 24, 37, 38, 14$.

a. $27, 28$

b. $30, 27$

c. $28, 29$

d. $28, 27$

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Algebra Essentials

A bag holds 6 red and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{3}{5}$

b. $\frac{8}{17}$

c. $\frac{2}{5}$

d. $\frac{7}{15}$

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